

React Practice Examples

→ List

Item itself as keys	Index as keys
<pre>import React from 'react'; const listitems = ['Tea', 'Ginger Tea', 'Masala Tea', 'Coffee', 'Boost']; function Lists() { return (<div> <h2>Hot Drinks</h2> {listitems.map((item) => (<li key={item}>{item}))} </div>); } export default Lists;</pre>	<pre>import React from 'react' const listitems=['Tea','Ginger Tea','Masala Tea','Coffee','Boost']; function Lists() { return (<div> <h2>Hot Drinks</h2> {listitems.map((item,index)=>(<li key={index}>{item}))} </div>) } export default Lists;</pre>

→ List items into the table

```
import React from 'react';

function ListToTable() {
  const dataList = [
    { id: 1, name: 'Alice', age: 25 },
    { id: 2, name: 'Bob', age: 30 },
    { id: 3, name: 'Charlie', age: 35 }
  ];

  return (
    <div>
      <h1>List to Table</h1>
      <table>
        <thead>
          <tr>
            <th>ID</th>
```

```

        <th>Name</th>
        <th>Age</th>
      </tr>
    </thead>
    <tbody>
      {dataList.map(item => (
        <tr key={item.id}>
          <td>{item.id}</td>
          <td>{item.name}</td>
          <td>{item.age}</td>
        </tr>
      ))}
    </tbody>
  </table>
</div>
);
}
export default ListToTable;

```

→ Background color change using useState:

```

import React, { useState } from 'react';

function BackgroundColorChanger() {
  const [bgColor, setBgColor] = useState('white');

  const changeBackgroundColor = () => {
    setBgColor(bgColor === 'white' ? 'lightblue' : 'white');
  };

  return (
    <div style={{ backgroundColor: bgColor, padding: '20px', textAlign: 'center' }}>
      <p>This background color changes!</p>
      <button onClick={changeBackgroundColor}>Change Background Color</button>
    </div>
  );
}

export default BackgroundColorChanger;

```

→ **Data is fetched and stored into a table:**

```
import React, { useState, useEffect } from 'react';

function FetchEvent() {
  const [data, setData] = useState([]);

  useEffect(() => {
    fetch('https://dummyjson.com/quotes')
      .then(response => response.json())
      .then(result => {
        setData(result.quotes);
      })
      .catch(() => {
        // Handle any errors here if needed
      });
  }, []);

  if (data.length === 0) return <p>No data available</p>;

  return (
    <div>
      <h1>Fetched Data</h1>
      <table>
        <thead>
          <tr>
            <th>ID</th>
            <th>Quote</th>
            <th>Author</th>
          </tr>
        </thead>
        <tbody>
          {data.map(item => (
            <tr key={item.id}>
              <td>{item.id}</td>
              <td>{item.quote}</td>
              <td>{item.author}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}
```

```
export default FetchEvent;
```

→ List items into the table

```
import React from 'react';

function ListToTable() {
  const dataList = [
    { id: 1, name: 'Alice', age: 25 },
    { id: 2, name: 'Bob', age: 30 },
    { id: 3, name: 'Charlie', age: 35 }
  ];

  return (
    <div>
      <h1>List to Table</h1>
      <table>
        <thead>
          <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Age</th>
          </tr>
        </thead>
        <tbody>
          {dataList.map(item => (
            <tr key={item.id}>
              <td>{item.id}</td>
              <td>{item.name}</td>
              <td>{item.age}</td>
            </tr>
          ))}
        </tbody>
      </table>
    </div>
  );
}

export default ListToTable;
```

→ **useReducer(Counter):**

```
import React, { useReducer } from 'react';
```

```
function reducer(state, action) {  
  switch (action.type) {  
    case 'increment':  
      return { count: state.count + 1 };  
    case 'decrement':  
      return { count: state.count - 1 };  
    default:  
      return state;  
  }  
}
```

```
function Counter() {  
  const [state, dispatch] = useReducer(reducer, { count: 0 });  
  
  return (  
    <div>  
      <p>Count: {state.count}</p>  
      <button onClick={() => dispatch({ type: 'increment' })}>+</button>  
      <button onClick={() => dispatch({ type: 'decrement' })}>-</button>  
    </div>  
  );  
}
```

```
export default Counter;
```

→ HOC Example

Counter 1	Counter 2	HOC
<pre>import React from 'react' import Update from './HOC'; const Counter1 = ({counting, increase}) => { return (<div> <h1>This is Counter-1</h1> <button onClick={increase}>Clicked me {counting} Times</button> </div>) } export default Update(Counter1)</pre>	<pre>import React from 'react' import Update from './HOC'; const Counter2 = ({counting, increase}) => { return (<div> <h1>This is Counter-2</h1> <button onClick={increase}>Clicked me {counting} Times</button> </div>) } export default Update(Counter2)</pre>	<pre>import React, {useState} from 'react'; const Update = (OriginalComponent) => { function NewComponent(){ const [count,setCount] = useState(0); const Up = () => { setCount(count + 1); }; return (<OriginalComponent counting={count} increase={Up} />) } return NewComponent; } export default Update</pre>